A Restaurant Recommendation Dialog System: Dialog Act Classification, Slot Filling and Rule-Based Reasoning

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Abstract

We build a restaurant recommendation dialog system for the DSTC 2 domain. Of a keyword baseline and three classifiers on bag-of-words (BoW) or frozen DistilBERT features, the BoW SVM classifies dialog acts of unseen utterances best (0.836 accuracy); the 95–99% on a random split mostly reflects that 84.6% of its test utterances occur in training. An eight-state transition function drives the dialog, with keyword slot extraction and a Levenshtein or semantic fallback that repair typos and paraphrases respectively. Six inference rules check additional requirements, with contradictions resolved in favour of the positive conclusion and explained to the user.

1. Introduction

A task-oriented dialog system has to work out what the user wants before it can respond: "is there anything else" asks for an alternative, whereas "cheap italian food" is a constraint to satisfy. Our system processes every user turn in a pipeline: a classifier labels the utterance with a *dialog act* (*inform*, *request*, ...), the dialog manager extracts preferences, looks up restaurants and selects the next system utterance, and a reasoning component checks additional requirements such as "romantic". Section 2 describes the pipeline, Section 3 evaluates the classifiers, slot extractors and dialogs, and Section 4 discusses leakage, error propagation, configurability and limitations.

2. System architecture

Dialog act classifier. The classifier maps a lowercased utterance to one of 15 dialog acts (Appendix A). A keyword baseline predicts the act of the highest-priority matching keyword (Table 3). We train a linear SVM, a decision tree (DT) and a one-layer MLP on two features: a BoW count vector and the mean of frozen DistilBERT token embeddings (Sanh et al., 2019). The dialog system uses the BoW SVM. This is a trade-off: on unseen utterances the rules have the higher balanced accuracy (0.727 vs. 0.657) because they cover rare acts, but the SVM has the higher accuracy (0.836 vs. 0.811) and macro-F1 (0.56 vs. 0.54), and the frequent acts (*inform*, *request*, *bye*) make up most turns. It also needs no GPU or torch at run time.

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Dialog manager. The core is a state transition function `transition(state, act, utterance)` that returns the next state and the system utterance. Its eight states are modelled on the DSTC 2 dialogs (Henderson et al., 2014) (Figures 1 and 2, Appendix B). System utterances are templates, separate from the logic. Preferences are used whatever act is predicted, so a misclassified "chinese food" is still understood.

Slot extraction. Keyword matching finds known values and patterns such as "{x} food" or "in the {x}", several slots per utterance, "any"/"don't care" and negations ("not chinese"). If a pattern word is unknown, a fallback chosen at run time maps it to the closest value: Levenshtein distance of at most two edits, or cosine similarity of DistilBERT embeddings above 0.60. A fallback value is always confirmed with the user, and a food type outside the database ("swedish") is rejected.

Lookup and reasoning. If several restaurants match, one is chosen at random and the others are kept for 'alternatives'. If none (or no more) match, the closest partial match is offered, dropping the price range first and then the area, never the food type. When at least two restaurants match, the user is asked for additional requirements and six rules derive *touristic*, *assigned seats*, *children* and *romantic* from food quality, crowdedness and length of stay (Table 4). A property only holds if a rule derives it.

3. Experiments & results

Classifiers. We evaluate on an 85/15 stratified random (*original*) split and a *grouped* split that keeps all copies of an utterance on the same side (Table 1). All learned models reach 95.5–98.7% on the original split but drop on the grouped split, where the BoW SVM is best (0.836) and the DT with DistilBERT drops most (to 0.601). BoW generalizes better than frozen embeddings for every classifier.

Table 1: Accuracy (Acc), balanced accuracy (BA) and macro F1. Best per column in bold.

		Original			**Grouped**		
Model	**Features**	Acc	BA	F1	Acc	BA	F1
Rule based	Rules	0.880	0.839	0.70	0.811	**0.727**	0.54
SVM	BoW	0.985	0.884	0.88	**0.836**	0.657	0.56
SVM	DistilBERT	**0.987**	0.886	**0.90**	0.804	0.646	0.54
MLP	BoW	0.985	**0.889**	**0.90**	0.829	0.648	**0.58**
MLP	DistilBERT	0.986	0.888	**0.90**	0.793	0.652	0.53
DT	BoW	0.975	0.849	0.86	0.800	0.581	0.52
DT	DistilBERT	0.955	0.730	0.75	0.601	0.295	0.20

Error analysis. Rare acts (*deny*, *ack*, *repeat*, *restart*, *reqmore*) mostly score below 0.5 F1. Three confusions dominate: *null* as *inform* (fragments such as "i want to find"), *thankyou* as *bye* (inconsistent labels for "thank you goodbye") and *affirm* as *inform* ("yes indian food"). Utterances with two acts are a frequent error for the learned models, because the data keeps only the first act as label: "hi im looking for moderately priced" is *inform*, but

the models follow "hi". On our 49 hand-written utterances (Appendix D), negation is hard for all models (0.17–0.65 accuracy).

Slot extractors. We compare both fallbacks on 41 hand-labelled phrases that keyword matching misses (Table 2). They fail in opposite ways. Levenshtein repairs every typo ("spenish") but has no notion of meaning: it turns "inexpensive" into *expensive*. DistilBERT maps "fancy" to *expensive* and "northern" to *north*, but no typos or dishes ("pizza"). It is 80 times slower and only usable after centring the embeddings, since frozen embeddings of single words are nearly identical. We use Levenshtein by default.

Table 2: Accuracy of the slot extraction fallbacks on 41 phrases (speed per utterance).

Fallback	Typos (16)	Paraphrases (16)	No match (9)	All (41)	ms
Levenshtein	**1.00**	0.00	**0.89**	**0.59**	**0.2**
Semantic	0.00	**0.44**	**0.89**	0.37	16

4. Discussion

Data leakage. Only about 4,800 of the 23,998 utterances are unique, and 84.6% of the original split's test utterances occur verbatim in its training data. Within that test set the learned models score 98–100% on these seen utterances and 84–92% on the unseen rest (71% for the DT with embeddings), so the original split mostly measures memorization. The grouped split, a separate evaluation without seen utterances, is the realistic estimate.

Error propagation. A wrong dialog act can send the dialog down the wrong path. Using preferences regardless of the predicted act, confirming uncertain values and answering requests classified as *inform* limit this.

Configurability. Two features can be switched (Appendix C): *reasoning transparency* (explain the inference chain or only name the restaurant) and *eager vs. patient* recommendation (suggest after the first preference or after all three). A user study can test whether explanations increase trust and whether fewer turns outweigh less fitting suggestions.

Limitations. Every utterance gets one act, the embeddings are not fine-tuned, the question order is fixed, and the rules use a closed world, so no restaurant can ever be "suitable for children".

5. Conclusion

We built a complete restaurant recommendation dialog system around a BoW SVM dialog act classifier, which generalizes best once data leakage is removed. Keyword slot extraction with a Levenshtein fallback handles typos, while the semantic fallback handles some paraphrases. Six inference rules with an explicit contradiction strategy explain each recommendation.

References

Matthew Henderson, Blaise Thomson, and Jason D. Williams. The second dialog state tracking challenge. In *Proceedings of the 15th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)*, pages 263–272, 2014.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

Appendix A. Data

Dialog acts. We use the dialog act annotations of the DSTC 2 restaurant domain (Henderson et al., 2014). Each line contains a dialog act label and a lowercased user utterance. The file has 23,998 utterances (about 4,800 unique) and 15 acts: *ack*, *affirm*, *bye*, *confirm*, *deny*, *hello*, *inform*, *negate*, *null*, *repeat*, *reqalts*, *reqmore*, *request*, *restart* and *thankyou*. The distribution is highly imbalanced: *inform* is by far the most frequent act, *reqmore* occurs 5 times. When an utterance had two acts (e.g. *thankyou* and *bye*), only the first is kept as its label.

Splits. The *original* split is a stratified random 85/15 split of all utterances. The *grouped* split assigns every unique utterance with all its copies to either the training or the test set (about 15% of the rows per class), so no test utterance occurs in training. All randomness uses seed 12345.

Dialogs and restaurants. The DSTC 2 dialog transcripts contain 3,235 dialogs; the most frequent user acts are *inform* (10,169), *request* (6,511), *bye* (3,367), *thankyou* (3,264), *reqalts* (1,754) and *null* (1,612). The restaurant database has 109 restaurants with area, food type, price range, phone, address and postcode; the extended version adds food quality, crowdedness and length of stay.

Appendix B. System details

Rule baseline. Table 3 lists the keywords and priorities (1 = highest; priority 9 is unused, only the order matters). Acts with specific keywords get priority over acts with general keywords; remaining ties were ordered by inspecting the data.

Learned classifiers. The BoW vocabulary is learned on the training set only, so unknown words are dropped. DistilBERT (`distilbert-base-uncased`) is frozen. The linear SVM uses `LinearSVC` (at most 5,000 iterations), the MLP one hidden layer of 100 units (at most 300 iterations) and the decision tree default settings. The DT was chosen for interpretability; it reaches 0.975 (BoW) and 0.955 (DistilBERT) on the original split but 0.800 and 0.601 on the grouped split. The MLP is close to the best model on both splits (0.985/0.829 with BoW). With DistilBERT the SVM is marginally best on the original split (0.987) but worse than with BoW on the grouped split (0.804).

Table 3: Rules of the keyword baseline.

Dialog act	Priority	Keywords
thankyou	1	thank you, thank, thanks, appreciate, cheers
hello	2	hi, hello, halo
bye	3	goodbye, cya, see you, bye
restart	4	start, over, reset
repeat	5	back, repeat, again
affirm	6	yes, ye, right, correct, affirmative
ack	7	okay, well, fine, kay
negate	8	no
deny	10	wrong, not
reqalts	11	about, else
reqmore	12	more
request	13	phone, address, price, post, what
inform	14	cheap, west, east, south, north, type, restaurant, moderate, moderately, priced, town, area, italian, indian, chinese, expensive, food, want, center, matter, care
confirm	15	is, it, that, does
null	16	unintelligible

State transition diagrams. Figure 1 shows the original diagram and Figure 2 the updated one with the additional requirements step. Every new piece of information passes the same decisions: an unknown food type is asked again (state 2), an uncertain value is confirmed (state 7), a missing preference is asked (state 2), and otherwise a restaurant is suggested (state 3) or the user is told that none matches (state 4). The update adds the decision "Additional requirements needed?" (at least two matching restaurants, not asked before and nothing recommended yet), state 8, and the option to add a requirement after a suggestion or a failed search. *Bye* and *restart* are accepted in every state; the diagrams show them where they occur in the data.

Slot extraction. Keywords are all values of the restaurant database plus a few synonyms ("center", "central" and "middle" for *centre*, "moderately" for *moderate*, "asian" for *asian oriental*), matched longest first so that "north american" is a food type and not an area.

Inference rules. Table 4 lists the rules and how many of the 109 restaurants they fire on. Only rules 5 and 6 can contradict each other in the data (37 restaurants); rules 1 and 2 would need a cheap romanian restaurant with good food, which does not exist. We considered three strategies for a contradiction: the negative conclusion wins (cautious, but excludes 37 restaurants, including the one expected in reference dialog 17), asking the user (one more turn), and the positive conclusion wins with a note to the user, which we chose because it keeps the candidate and stays transparent.

Table 4: Inference rules and the number of restaurants they fire on.

Rule	If	Then	Reason given to the user	Fires
1	cheap and good food	touristic	it is cheap and has good food	12
2	romanian	not touristic	romanian food is unfamiliar to tourists	1
3	busy	assigned seats	the waiter decides where you sit	62
4	long stay	not for children	you stay a long time	64
5	busy	not romantic	it is busy	62
6	long stay	romantic	it allows you to stay for a long time	64

Appendix C. Configurability

Both features are command line options of `dialog_manager.py`; everything else stays the same, so two versions can be compared in a user study.

Reasoning transparency (`--transparency on|off`). With transparency on, the recommendation explains every additional requirement with the rule that derived it and mentions a disagreeing rule. With transparency off the same restaurant is recommended as "frankie and bennys is a nice restaurant serving italian food in the south of town in the expensive price range."

Eager vs. patient recommendation (`--recommendation-mode`). The patient system asks for food type, area and price range, then for additional requirements, before it suggests a restaurant. The eager system suggests a restaurant as soon as one preference is known: after "cheap chinese food" it answers "charlie chan is a nice restaurant serving chinese food in the centre of town in the cheap price range" instead of asking for the area.

Appendix D. Experiments and Results

Difficult cases. Table 5 gives the accuracy of all models on our own utterances, which are listed with their labels in Table 6.

Table 5: Accuracy on our own difficult cases (emb = DistilBERT features).

Set	Rule	SVM	MLP	DT	SVM emb	MLP emb	DT emb
Negation (23)	0.52	0.52	**0.65**	0.57	0.61	0.57	0.17
Noisy (26)	0.65	**0.73**	0.69	0.58	**0.73**	0.69	0.38

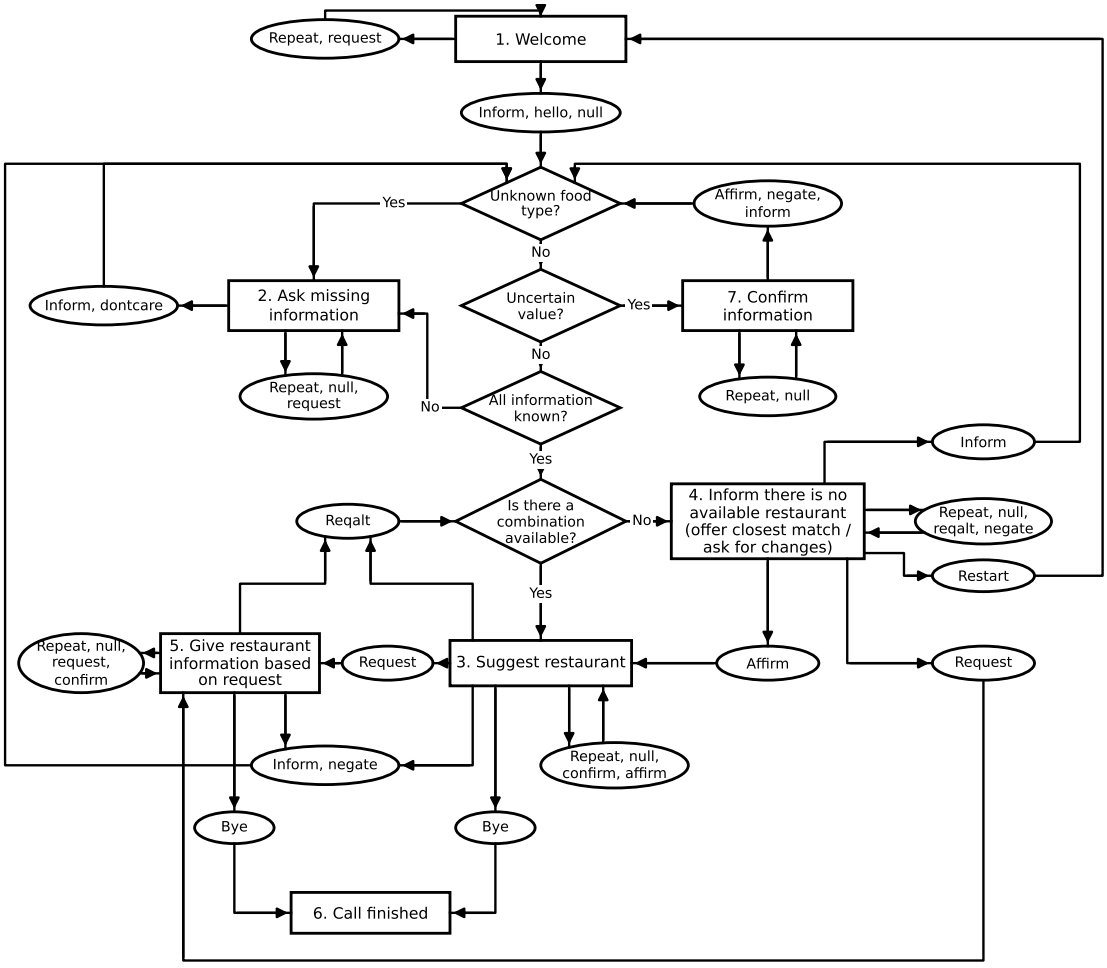


Figure 1: Original state transition diagram. Rectangles are system utterances, ovals user dialog acts and diamonds decisions.

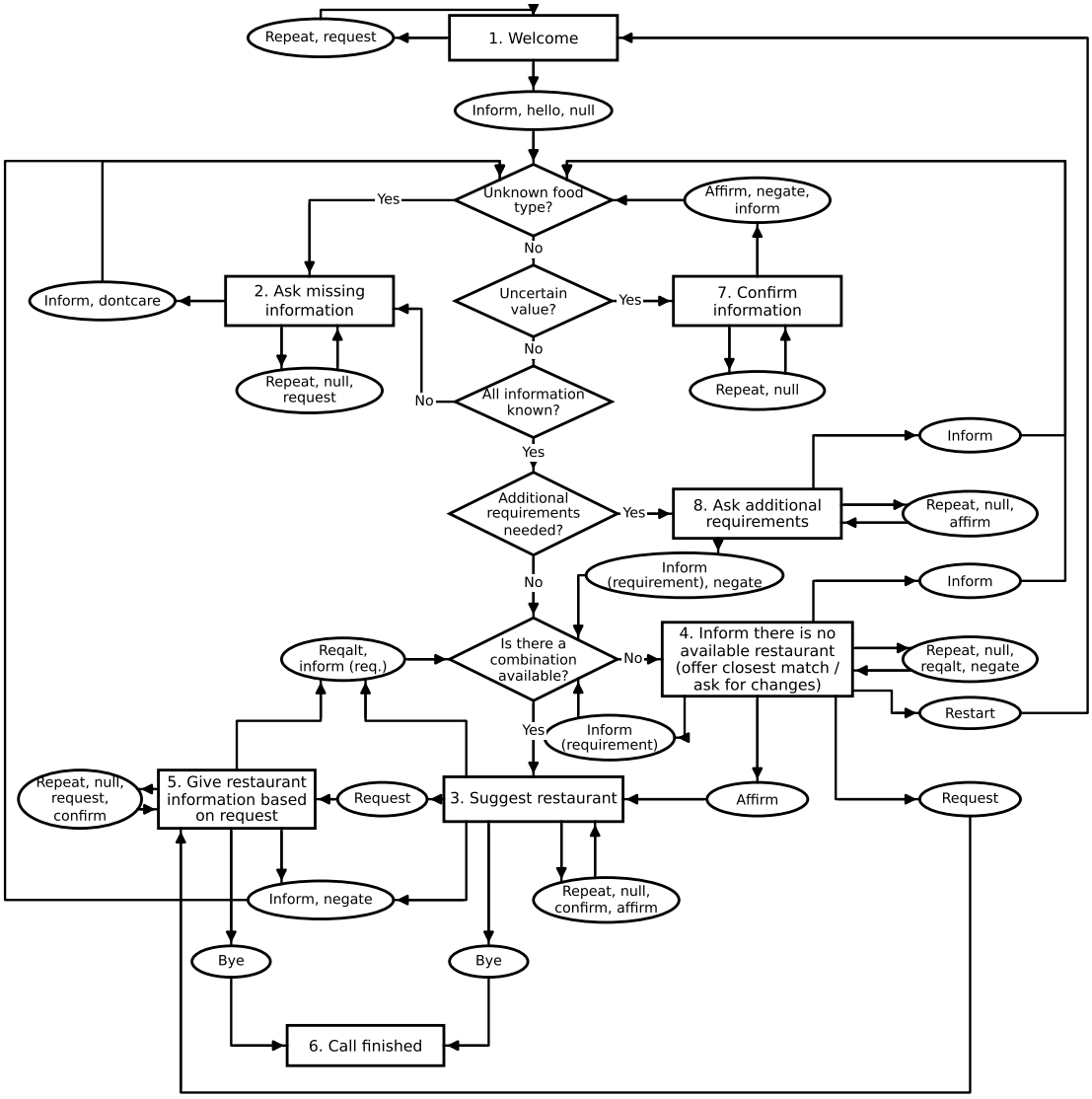


Figure 2: Updated state transition diagram with the additional requirements step (decision "Additional requirements needed?" and state 8).

Table 6: Our difficult utterances with their gold labels.

Negation (23)		**Noisy (26)**	
deny	i dont want an expensive restaurant	inform	uh i want um cheap food
deny	i do not want chinese food	inform	i i want a a restaurant in the in the south
deny	not in the north part of town	inform	im looking for a resturant that serves italien food
deny	i dont want it to be cheap	inform	cheep restaurant nort part of town
deny	not italian food	inform	um moderetly uh priced
deny	i dont want to eat in the center	inform	a restaurant serving uh tv_noise korean food
deny	not the expensive one	inform	any any area is fine uh
deny	i would not like spanish food	inform	spanish spanish food please
inform	i really dont care what area	request	whats the uh the phone number
inform	it does not matter what kind of food	request	adress please
inform	i dont mind the price range	request	can i get the post uh the post code
negate	no i want an expensive restaurant	request	what is the the adres of it
negate	no i said the east part of town	thankyou	thank thank you
negate	no thank you that is all	thankyou	thnks
negate	no i did not say that	bye	goodby
confirm	isnt it in the south part of town	bye	uh bye bye
confirm	doesnt it serve thai food	affirm	yeah yeah
confirm	is it not moderately priced	affirm	yea that is right uh
reqalts	i dont like that one is there anything else	negate	no no no uh
reqalts	not that one how about another restaurant	repeat	wha what did you say
request	i dont need the address just the phone number	repeat	sorry can you uh repeat
request	i dont know the postcode what is it	reqalts	uh anything uh else
thankyou	thank you but i dont need anything else	reqalts	how bout uh chinese
		confirm	is is it in the uh center
		restart	uh start uh start over
		hello	helo

Data leakage. The 84.6% is the share of test rows of the original split whose exact (lowercased) utterance also occurs among the training rows. Accuracy on the seen and unseen parts is computed within that same test set; the grouped split is a separate evaluation in which no test utterance is seen.

Example dialogs. In the first dialog a typo is confirmed before it is used; in the second a failed search offers the closest partial match.

Dialog 1: typo confirmation

U: I want spenish food in the centre
S: I did not recognize 'spenish'. Did you mean spanish?
U: yes
S: Would you like something in the cheap, moderate, or expensive price range?

Dialog 2: partial match

U: indian food in the west
S: Would you like something in the cheap, moderate, or expensive price range?
U: moderate
S: I'm sorry, there is no restaurant serving indian food in the west of town in the moderate price range. The closest match is india house, a restaurant serving indian food in the west of town, but it is in the expensive price range instead of moderate. Would you like that one, or would you like to change your preferences?
U: yes
S: Okay, india house it is. Can I help you with anything else?

Appendix E. Contributions

The work was split by component so that everyone worked on both parts; the hours are roughly equal.

Table 7: Contributions per group member.

Member	Tasks	Hours	Reflection
Junaid Malik	MLP, slot extraction, dialog manager, testing the code	30	
Abischan Manimaran	Rule-based classifier, state transition diagrams, configurability (eager/patient), testing the dialogs Decision tree, restaurant lookup,	30	
Milan Schouten	report	30	
Luca Spermezan	Support vector machine, the six inference rules and contradiction handling, large part of the report	30	

Appendix F. AI disclosure statement

We, Junaid Malik (7077424), Abischan Manimaran (4644603), Milan Schouten (4453239), and Luca Spermezan (1931415) hereby confirm that the submitted work for the course *Methods in AI research* is our own work. We declare that we have only used AI tools as permitted for this work.

Table 8: Overview of AI Tools Used

Name of the AI tool	Purpose of using the AI tool
ChatGPT	Grammar check

We acknowledge our responsibility as students/learners to thoroughly check all output and content produced by AI tools, and we accept full responsibility for its accuracy and validity.

Names and signatures of all students:

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Date: October 11, 2026